

Zin Win Thu

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Research Summary

Ph.D. researcher in Aerospace Engineering specializing in **Generative AI for aircraft design**, **probabilistic surrogate modeling (MDN)**, **uncertainty-based optimization**, and **hybrid-electric aircraft conceptual design**. Lead developer of the **CADE** multidisciplinary aircraft design environment, a framework that supports the entire workflow from early-stage design generation to multidisciplinary design optimization, integrating geometry, aerodynamics, weight estimation, propulsion, performance analysis, and optimization.

Core Expertise

- Generative AI (VAE), probabilistic modeling (BNN, MDN)
- Robust design optimization, Reliability-based design optimization, Possibility-based design optimization
- Multidisciplinary design optimization (MDO)
- Hybrid/electric aircraft performance and sizing for UAV, UAM and RAM.

Education

Ph.D.–M.Sc., Aerospace Information Engineering, Konkuk University (2020–2025)

Thesis: GenAI Optimization Approach for Large Scale Aircraft Design under Uncertainty

B.E., Aerospace Engineering, Myanmar Aerospace Engineering University (2012–2019)

Thesis: Design and Construction of Mini Jet Engine

Selected Publications

- Under review: *Journal of Aircraft*, 2025 — Generative AI + probabilistic surrogate modeling for airfoil design.
- Thu et al., *IEEE Access*, 2025 — Hybrid-electric aircraft sizing using PBDO.
- Thu et al., *IEEE TTE*, 2024 — AAM propulsion system framework.
- Thu et al., *Applied Sciences*, 2022 — UAM scheduling optimization.

Selected Research Projects

CADE – Comprehensive Aircraft Design Environment

A framework that supports the entire workflow from early-stage design generation to multidisciplinary design optimization, integrating geometry, aerodynamics, weight, propulsion, performance analysis, and optimization.

Concept Development of Next Generation Regional Air Mobility Aircraft

Development of hybrid electric aircraft for regional operations.

UAM Operational Digital Twin

Real-time multi-vehicle mission planning and AAM operations analysis.

Industry Experience

- Air KBZ Airline — Aircraft maintenance trainee (2018)
- Myanmar National Airlines — Industrial trainee (2016)

Volunteering

IWIK — English teacher for orphans; community event manager (2023–Present)

Skills

Python, MATLAB, L^AT_EX

Reference

Prof. Jae-Woo Lee, Konkuk University — jwlee@konkuk.ac.kr

Prof. Maxim Tyan, New Uzbekistan University - m.tyan@newuu.uz